Distributed Pound-Drever-Hall Technique for Fast Brillouin Optical Time Domain Analysis

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***Abstract*—The Pound-Drever-Hall (PDH) technique is a powerful method for laser frequency stabilization, and a PDH loop can also act as a single-channel sensor with high resolution and fast response speed. In this work, we expand the PDH technique into a time-domain distributed form, accomplishing a fast Brillouin optical time domain analysis (BOTDA) scheme which measures the distribution of Brillouin frequency shifts along optical fiber without frequency scanning. Approaches of phase modulated probe light, synchronous demodulation for PDH error, arbitrarily accessing in single interrogation, and distributed form feedback control are employed in the system. The scheme achieves fast measurement speed and large dynamic range at the same time, as well as good response linearity and high tolerance to optical power fluctuation. In the demonstration experiment, sensing resolution of 0.4 MHz and measurement range of 300 MHz are realized, under the condition of 25 km single mode communication fiber, 5 m spatial resolution and 0.24 s measurement time (400 times of averaging).**

***Index Terms*—Optical fiber sensor, Pound-Drever-Hall technique, stimulated Brillouin scattering.**

I. Introduction

THE Pound-Drever-Hall (PDH) technique [1], [2] is a widely known method for laser frequency stabilization, by locking the laser to a stable optical resonator with feedback controller. In addition, a PDH loop can also serve as a high-precision and fast-response sensor when the laser is considered stable enough and the optical cavity acts as the sensitive element, and such configuration has been adopted in optical systems ranging from the extremely high resolution laser interferometer gravitational-wave observatory (LIGO) [3], [4], to a variety of optical sensors relying on precise laser frequency discrimination [5], [6]. Brillouin optical time domain analysis (BOTDA) [7] is an important distributed optical fiber sensing technology, which employs stimulated Brillouin scattering (SBS) of optical fiber to acquire the distribution of strain or temperature along the fiber. Due to the advantages of intrinsic absolute readout, long-distance and high-density measurement, immunity to electromagnetic interference, BOTDA has been widely applied in the scenarios such as structural health monitoring [8], hydrology [9] and pipeline leakages [10].

The original BOTDA scheme [11] depends on frequency scanning with wide range and high granularity to get the Brillouin gain spectrum (BGS) along the fiber, which limits the measurement speed of BOTDA systems, especially for those with long-range sensing fibers exceeding 10 km where averaging is mandatory to achieve an adequate signal-to-noise ratio (SNR) [12]. Therefore, the fast BOTDA scheme, i.e., method to escape the redundant frequency scanning, is one of the most studied aspects of BOTDA technology. Works in [12], [13] proposed the slope-assisted (SA) BOTDA directly using the intensity of the SBS response as the sensor's readout, while facing an optical power fluctuation dependent noise. To solve such imperfection, fast BOTDA schemes based on single-sideband-modulated probe light [14], phase-modulated probe light (with different demodulation strategies) [15], [16], [17], and dual frequency probe light [18] were proposed to avoid the impact of power fluctuation. Methods above have another common restriction, which is the measurement range is limited to tens of megahertz determined by the intrinsic SBS bandwidth. Further optimizations such as the multi-slope assisted BOTDA [19], the Brillouin phase-to-gain ratio (BPGR) scheme [20], the combination of multi-slope assisted and BPGR [21], and the BGS shape manipulation scheme [22] enlarge the measurement range of SA BOTDA to 100 MHz level, but their large signal responses are not perfectly linear and hence measurement results manifest distortions to some degree.

Other fast BOTDA schemes in recent year include using rapidly-repeated chirping probe light [23], [24] or frequency comb probe light [25], [26]. By employing such periodically synthesized probe light, these schemes achieved excellent performance, while in the authors' opinion there would be still some incompleteness about the time-consuming nonlinear curve fitting for readouts, and the excess modulation bandwidth adverse to the system SNR (more detailed discussion will be given later). Another key approach proposed to enlarge the measurement range of scanning-free BOTDA, is the usage of probe light with arbitrary varying frequency, *tracking* the real Brillouin frequency shift (BFS) everywhere along the sensing fiber [12], [27], instead of probe light with fixed center frequency. The concept was first mentioned in [12] (probe light with *preset* arbitrarily-varying frequency was demonstrated), and

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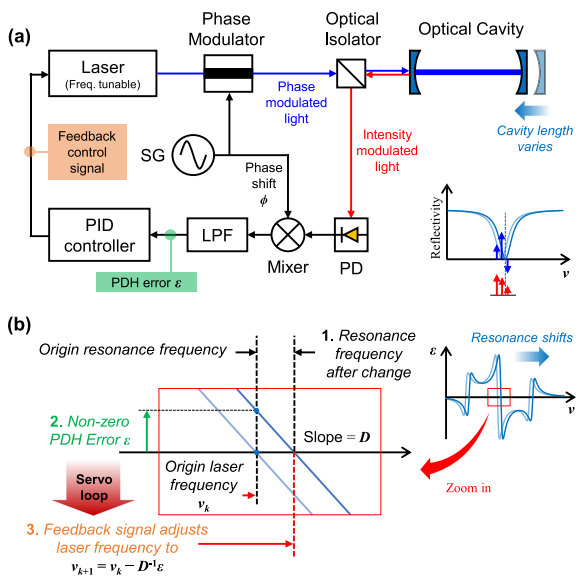

Fig. 1. Sketch of the origin PDH technique. (a) System configuration of a typical PDH loop [2]. (b) The PDH error curve and the simplified discrete feedback process.

implemented (self-adjusting during execution) by Yang et al. in the closed-loop control (CC) BOTDA scheme [27]. And to solve the issue of optical power fluctuation noise, an evolutionary scheme, the gain-ratio tracking BOTDA, was proposed in [28] employing dual frequency probe light similar to the concept in [18]. While, the gain-ratio tracking BOTDA is still affected by the frequency domain probe power fluctuation to some degree (as will be discussed later).

According to the introduction above, it can be found that the PDH technique and the fast BOTDA share many technical concepts, such as instantaneous frequency discrimination, Lorentzian spectra (for both optical cavity and BGS), periodically modulated "probe" light, and feedback control. But it is obvious that the PDH technique cannot be directly introduced into BOTDA, since a PDH loop only supports one optical cavity (one sensing channel). So, in this work, we extend the origin PDH technique into a distributed form, which fits well with the BOTDA system, named the distributed Pound-Drever-Hall (DPDH) BOTDA. The optical system is based on the classical BOTDA configuration and an additional optical phase modulator (PM) in the probe branch. Besides, the system control is accomplished by a digital signal platform which provides sufficient flexibility in signal generation, acquisition, and processing. In the demonstrational experiment, sensing resolution of 0.4 MHz (at the far end of fiber) and measurement range of 300 MHz are achieved over 25 km fiber, with spatial resolution of 5 m and measurement time of 0.24 s (400 times averaging).

II. Principle

A. A Brief Review of Origin PDH Technique

For ease of understanding, a brief review of the original PDH technique is given here. Fig. 1(a) illustrates a typical PDH loop [2], which mainly contains a frequency tunable laser, a PM, an optical cavity, a synchronous demodulator (mixer and low pass filter) and a feedback controller. An important concept is that the combination of sinusoidal phase-modulated incident light and the synchronous demodulation on the intensity ofreflected light establishes a signed, linear (near the resonance center), and immediate frequency mismatch discrimination between the laser and the resonance of optical cavity. The output signal of synchronous demodulator is the PDH error ε, which quantitatively characterizes the frequency mismatch and is given by [2]

$$\varepsilon(\nu) = 2\sqrt{P_{\mathrm{ca}}P_{\mathrm{sb}}}\{\mathrm{Re}[\boldsymbol{f}(\nu)]\cos\phi + \mathrm{Im}[\boldsymbol{f}(\nu)]\sin\phi\},$$

$$\boldsymbol{f}(\nu) = \boldsymbol{R}(\nu)\boldsymbol{R}^*(\nu + f_{\mathrm{m}}) - \boldsymbol{R}^*(\nu)\boldsymbol{R}(\nu - f_{\mathrm{m}}),$$

$$\boldsymbol{R}(\nu) = \frac{r[\exp(2\pi j\nu/\nu_{\mathrm{fsr}}) - 1]}{1 - r^2\exp(2\pi j\nu/\nu_{\mathrm{fsr}})} \tag{1}$$

where ν is the laser frequency, P_{ca} and P_{sb} are the power of carrier and first-order sidebands of the phase-modulated incident laser, f_{m} is the phase modulation frequency, r and ν_{fsr} are the mirrors' amplitude reflection coefficient and the free spectral range of the optical cavity respectively. Term $\boldsymbol{R}(\nu)$ represents the complex amplitude reflection coefficient of the cavity (corresponding to an approximate Lorentzian lineshape $|\boldsymbol{R}(\nu)|^2$), which is conjugate symmetric centered at the resonance (ν equaling to integral times of ν_{fsr}). Hence, the intermediate function $\boldsymbol{f}(\nu)$ is intrinsically asymmetric with respect to resonance centers, and so is the error $\varepsilon(\nu)$. The parameter ϕ represents a certain phase shift applied on the sinusoidal modulation signal before entering the frequency mixer. To maximize the slope of the error signal, ϕ should take 0 when $f_{\mathrm{m}} \ll \Delta\nu$, or takes 90° when $f_{\mathrm{m}} \gg \Delta\nu$, or can be empirically determined when f_{m} and $\Delta\nu$ are close ($\Delta\nu = (1 - r^2)\nu_{\mathrm{fsr}}/\pi$ is the resonance width of the cavity) [2]. As shown in Fig. 1(b), the error signal provides a direct frequency mismatch discrimination near the resonance center. When $\phi \approx 90°$, the PDH frequency discriminant, i.e., the slope of $\varepsilon(\nu)$ near resonance is [2]

$$D = -8\sqrt{P_{\mathrm{ca}}P_{\mathrm{sb}}}/\Delta\nu. \tag{2}$$

Applying feedback control on frequency tunable laser according to the PDH error forms a PDH loop. For a free-running laser, a stable resonator, and a high bandwidth analog feedback controller, the laser phase noise can be successfully suppressed. While, for PDH loop for sensing purpose and the absolute frequency (rather than the frequency mismatch) is interested, a simplified discrete, pure-integral feedback control model is also effective [6] (and fits the DPDH BOTDA scheme later). The feedback control model is illustrated in Fig. 1, and can be described by the straightforward iterative process below

$$\nu_{k+1} = \nu_k - D^{-1}\cdot\bar{\varepsilon}_k, \quad k = 0, 1, \cdots \tag{3}$$

where ν_k is the adjusted laser frequency at the beginning of the k-th iteration, and $\bar{\varepsilon}_k$ is the average value of error signal within one iteration interval after the k-th adjustment. Such a pure-integral feedback process cyclically adjusts the laser frequency

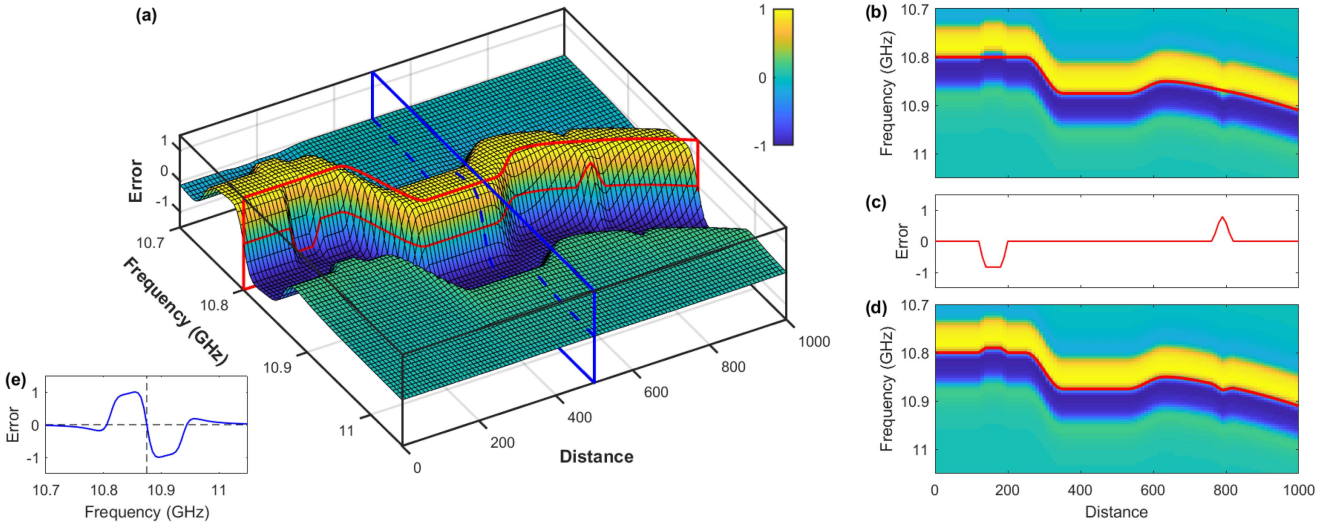

Fig. 2. A conceptual illustration of the distributed PDH technique for BOTDA. (a) The 3-D error surface of sensing fiber with arbitrary BFS distribution (Fiber attenuation is not considered). Note that the arbitrary accessing probe light (depicted by the red box) is aligned with the actual BFS distribution almost everywhere, except the locations near distance = 180 and 800 which represents the situation that BFS changes suddenly occur. (b) Top view of sub-figure (a) representing the initial arbitrary accessing probe light. (c) Side view of the red box in sub-figure (a), in which the red curve is the acquired PDH error distribution. Note that non-zero PDH errors occur near distance = 180 and 800. (d) The adjusted arbitrary accessing probe light (the red curve) according to the PDH error distribution. Now it is aligned with the actual BFS distribution everywhere. (e) Side view of the blue box in sub-figure (a) demonstrating the PDH error over frequency.

to track the resonance, so the sequence $\{\nu_k\}$ itself is the sensing result. Note that the averaging of error signal within one iteration interval vanishes the dynamic information in the error signal above the feedback bandwidth, which is meaningful for PDH-loop-based dynamic sensors [5]. In this work we ignore such dynamic results for simplicity, and focus on the adjusted laser frequency results, which covers the band from DC to half the iteration rate, and the measurement range is as large as the laser's tuning range (rather than the relatively small linear region near resonance in the error vs. frequency curve).

B. Distributed PDH Technique for BOTDA

BOTDA technology employs the interaction between the forward-propagating pulsed pump light and the counter-propagating continuous probe light (Stokes light for gain-type BOTDA), i.e., the SBS in the sensing fiber, to acquire the BFS distribution. The response of probe light (the ratio of probe light after/before SBS) is characterized by the BGS which is also in approximate Lorentzian lineshape. So, it is natural that PDH error of SBS can be obtained by using phase-modulated probe light in combination with synchronous demodulation. A slight difference is that the frequency argument of PDH error of SBS is frequency difference between pump and probe light (denoted by ν'), rather than the laser frequency ν in (1). Moreover, since the BOTDA is a distributed sensor and each segment of sensing fiber has independent BFS, the PDH error of BOTDA has another distance dimension and is denoted as $\varepsilon(\nu', z)$ where z is the distance argument satisfying the time-of-flight principle.

In order to fully exploit the PDH technique into BOTDA system, the interrogation process and the feedback control process should also be expanded into a distributed form. Once a pump pulse enters the near end of sensing fiber, its center frequency is no longer controllable of course. While, the continuous probe light entering the fiber from the far end provides a chance that the frequency difference ν' between pump and probe light when SBS occurs at any position z can be different and independent, just by composing a probe wave with time-varying frequency. This approach was also described in detail in [12], [27], and for PDH system in distributed form, the only necessary modifications are applying sinusoidal phase modulation on the probe light at the same time, and introducing synchronous demodulation after the photodetection. Fig. 2(a) demonstrates such arbitrarily accessing interrogation process, where the error surface $\varepsilon(\nu', z)$ representing a segment of sensing fiber with non-constant BFS $\nu_{\mathrm{B}}(z)$ is intersected by a curve surface $\nu' = \nu'_{\mathrm{init}}(z)$ (depicted by the red box) corresponding to an initially arranged probe light with time-varying frequency.

In Fig. 2(a), the intersection of the two curves (marked in thin red line) represents the PDH error $\varepsilon(z)$ which discriminates the mismatch between $\nu'_{\mathrm{init}}(z)$ and $\nu_{\mathrm{B}}(z)$. It is natural that we can get a new accessing frequency $\nu'_{\mathrm{adj}}(z)$ aligning to the real BFS distribution $\nu_{\mathrm{B}}(z)$, by adjusting the $\nu'_{\mathrm{init}}(z)$ with $\varepsilon(z)$ like (3) but in an extended parallel, one-to-one pattern (Fig. 2(b) to (d) illustrates such an adjustment). By cyclically repeating the adjustment above, a distributed form feedback control process can be established, which makes the probe light always track the real BFS distribution along the sensing fiber.

Theoretical expression of $\varepsilon(\nu', z)$ can be derived from the analytical models on BOTDA contributed in [29], in combination with the PDH error (1) given in [2]. For a general condition in which the frequency ν' and the distance z are completely arbitrary, and the power consumption of pump pulse is *not* negligible, the expression of pump power at any z will be in a redundant form, i.e., contains a non-simplifiable integration from 0 to z corresponding to the accumulative power consumption by SBS.

However, under a reasonable restriction that the ν' at any z is nearly aligned with the actual BFS distribution $\nu_{\mathrm{B}}(z)$, which is the situation when the non-local effect is most significant as investigated in [29], as well as the actual situation when a DPDH BOTDA system is in operation, a closed-form expression of $\varepsilon(\nu', z)$ exists.

For clarity, the polarization fading effect is not considered. The (10) in [29] gives the basic power transfer equation for the Brillouin interaction when the pump light is always under a maximum consumption, i.e, the ν' at any z is always the same to $\nu_{\mathrm{B}}(z)$. This equation can be modified into a form that the probe light is phase modulated and mainly contains the carrier and the $\pm$1st order sidebands, which is

$$\begin{cases} \mathrm{d}P_{\mathrm{P}}(z) = -\alpha P_{\mathrm{P}}(z)\mathrm{d}z \\ \qquad\qquad -g_{\mathrm{B}}A_{\mathrm{eff}}^{-1}P_{\mathrm{P}}(z)[P_{\mathrm{ca}}(z) + \frac{2\Delta_{\mathrm{B}}^2}{\Delta_{\mathrm{B}}^2+4f_{\mathrm{m}}^2}P_{\mathrm{sb}}(z)]\mathrm{d}z, \\ P_{\mathrm{ca}}(z) = P_{\mathrm{ica}}e^{-\alpha(L-z)}, \quad P_{\mathrm{sb}}(z) = P_{\mathrm{isb}}e^{-\alpha(L-z)}, \\ P_{\mathrm{P}}(0) = P_{\mathrm{iP}}, \quad z \in [0, L] \end{cases} \tag{4}$$

where $P_{\mathrm{P}}(z)$ is the pump power distribution, α is the fiber attenuation, g_{B} is the peak value of the Brillouin gain, A_{eff} is the effective mode area of fiber, Δ_{B} is the full width at half maximum (FWHM) of the BGS, f_{m} is the phase modulation frequency, L is the total length of fiber, $P_{\mathrm{ca}}(z)$ and $P_{\mathrm{sb}}(z)$ are the power distributions of carrier and first order sidebands of the probe light respectively, P_{iP}, P_{ica} and P_{isb} are the input power of pump, probe carrier and probe sidebands, respectively. P_{ica} and P_{isb} satisfy $P_{\mathrm{ica}} + 2P_{\mathrm{isb}} = P_{\mathrm{iS}}$ and P_{iS} is the total power of the input Stokes probe light. The differential equation in (4) has a closed-form solution (similar to the (11) in [29]) which is:

$$\begin{aligned} &P_{\mathrm{P}}(z) = P_{\mathrm{iP}}A(z), \quad \text{with} \\ &A(z) = \exp\left\{-\alpha z - g_{\mathrm{B}}A_{\mathrm{eff}}^{-1}\left[P_{\mathrm{ica}} + \frac{2\Delta_{\mathrm{B}}^2}{\Delta_{\mathrm{B}}^2+4f_{\mathrm{m}}^2}P_{\mathrm{isb}}\right]z_{\mathrm{eq}}\right\} \\ &\text{and} \quad z_{\mathrm{eq}} = \frac{e^{-\alpha(L-z)} - e^{-\alpha L}}{\alpha} \end{aligned} \tag{5}$$

where the term $A(z)$ represents a normalized amplitude along z, which contains the fiber attenuation, and the accumulative power consumption of the pump pulse during its propagation. Note that an equivalent distance z_{eq} as defined in (5) is used to denote the accumulative pump power consumption.

With the pump power distribution in (5), the local SBS response in complex amplitude at any z can be expressed as (under the small gain condition):

$$\begin{aligned} &\boldsymbol{G}_{\mathrm{local}}(\delta_\nu, z) = \exp\left[\frac{g_{\mathrm{B}}A_{\mathrm{eff}}^{-1}P_{\mathrm{P}}(z)l}{2(1+2j\delta_\nu/\Delta_{\mathrm{B}})}\right] = A(z)\boldsymbol{G}_{\mathrm{u}}(\delta_\nu), \\ &\text{with} \quad \boldsymbol{G}_{\mathrm{u}}(\delta_\nu) = \exp\left[\frac{g_{\mathrm{B}}A_{\mathrm{eff}}^{-1}P_{\mathrm{iP}}l}{2(1+2j\delta_\nu/\Delta_{\mathrm{B}})}\right] \\ &\qquad\qquad \approx 1 + \frac{g_{\mathrm{B}}A_{\mathrm{eff}}^{-1}P_{\mathrm{iP}}l}{2\left(1+2j\delta_\nu/\Delta_{\mathrm{B}}\right)} \end{aligned} \tag{6}$$

where $\delta_\nu = \nu' - \nu_{\mathrm{B}}$ represents the difference in frequency between pump and probe ν' and the actual BFS ν_{B}, and l is the SBS interaction length (and the nominal spatial resolution of BOTDA) equaling to $c_0 t_{\mathrm{p}}/2n_{\mathrm{g}}$ where c_0 is vacuum light speed,

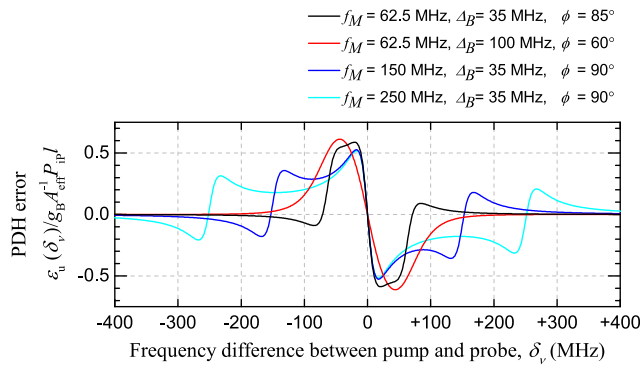

Fig. 3. Theoretical PDH error curves of SBS in BOTDA by (8) under different combinations of f_{M}, Δ_{B} and ϕ.

t_{p} is the pump pulse duration, and n_{g} is the group refractive index of fiber. The SBS response $\boldsymbol{G}_{\mathrm{local}}(\delta_\nu, z)$ can be written into the product of $A(z)$ and $\boldsymbol{G}_{\mathrm{u}}(\delta_\nu)$, and $\boldsymbol{G}_{\mathrm{u}}(\delta_\nu)$ represents a unified SBS response corresponds to the Lorentzian SBS response in power $g(\delta_\nu) = |\boldsymbol{G}_{\mathrm{u}}(\delta_\nu)|^2 \approx 1 + g_{\mathrm{B}}A_{\mathrm{eff}}^{-1}P_{\mathrm{iP}}l/[1+(2\delta_\nu/\Delta_{\mathrm{B}})^2]$ when $g_{\mathrm{B}}A_{\mathrm{eff}}^{-1}P_{\mathrm{iP}}l \ll 1$ [30]. Since the $\boldsymbol{G}_{\mathrm{u}}(\delta_\nu)$ is conjugate symmetric at $\delta_\nu = 0$, the distributed form of the intermediate function $\boldsymbol{f}(\nu)$ in (1) is intrinsically asymmetric, and can be expressed as:

$$\begin{aligned} &\boldsymbol{f}_{\mathrm{local}}(\delta_\nu, z) = A(z)\boldsymbol{f}_{\mathrm{u}}(\delta_\nu), \quad \text{with} \\ &\boldsymbol{f}_{\mathrm{u}}(\delta_\nu) = \boldsymbol{G}_{\mathrm{u}}(\delta_\nu)\boldsymbol{G}_{\mathrm{u}}^*(\delta_\nu + f_{\mathrm{m}}) - \boldsymbol{G}_{\mathrm{u}}^*(\delta_\nu)\boldsymbol{G}_{\mathrm{u}}(\delta_\nu - f_{\mathrm{m}}) \end{aligned} \tag{7}$$

where $\boldsymbol{f}_{\mathrm{u}}(\delta_\nu)$ represents a unified intermediate function. Then, by employing the relationship between $\boldsymbol{f}(\nu)$ and $\varepsilon(\nu)$ in (1) and taking the fiber attenuation until the SBS light reaching the photodetector into account, we can get the theoretical expression of the PDH error in BOTDA which is

$$\begin{aligned} \varepsilon(\nu', z) &= e^{-\alpha z} \cdot 2\sqrt{P_{\mathrm{ca}}(z)P_{\mathrm{sb}}(z)}A(z)\varepsilon_{\mathrm{u}}[\nu' - \nu_{\mathrm{B}}(z)] \\ &= 2\sqrt{P_{\mathrm{ica}}P_{\mathrm{isb}}}e^{-\alpha L}A(z)\varepsilon_{\mathrm{u}}[\nu' - \nu_{\mathrm{B}}(z)], \\ \text{with} \quad \varepsilon_{\mathrm{u}}(\delta_\nu) &= \mathrm{Re}[\boldsymbol{f}_{\mathrm{u}}(\delta_\nu)]\cos\phi + \mathrm{Im}[\boldsymbol{f}_{\mathrm{u}}(\delta_\nu)]\sin\phi \end{aligned} \tag{8}$$

where the principle of determining the value of parameter ϕ is identical to that as described before in (1). The error $\varepsilon(\nu', z)$ is the product of three terms. The first term $2\sqrt{P_{\mathrm{ica}}P_{\mathrm{isb}}}e^{-\alpha L}$ is a global scaling constant. The second term $A(z)$ determines the relative amplitude of the error signal along z as defined in (5). The third term $\varepsilon_{\mathrm{u}}(\delta_\nu)$ represents a unified PDH error of SBS which is intrinsically asymmetric and irrelevant to z. Fig. 3 shows the curves of $\varepsilon_{\mathrm{u}}(\delta_\nu)/g_{\mathrm{B}}A_{\mathrm{eff}}^{-1}P_{\mathrm{p}}l$ under different combination of f_{m} and Δ_{B} with appropriate ϕ. Lastly, similar to (2), when f_{m} is greater than Δ_{B} and $\phi \approx 90°$, the PDH frequency discriminant in distributed form for BOTDA can be written as

$$\begin{aligned} D(z) &\equiv \frac{\mathrm{d}\varepsilon(\nu', z)}{\mathrm{d}\nu'}\Big|_{\nu'=\nu_{\mathrm{B}}(z) \text{ or } \delta_\nu=0} \\ &\approx -4g_{\mathrm{B}}A_{\mathrm{eff}}^{-1}P_{\mathrm{iP}}l\sqrt{P_{\mathrm{ica}}P_{\mathrm{isb}}}e^{-\alpha L}A(z)/\Delta_{\mathrm{B}}. \end{aligned} \tag{9}$$

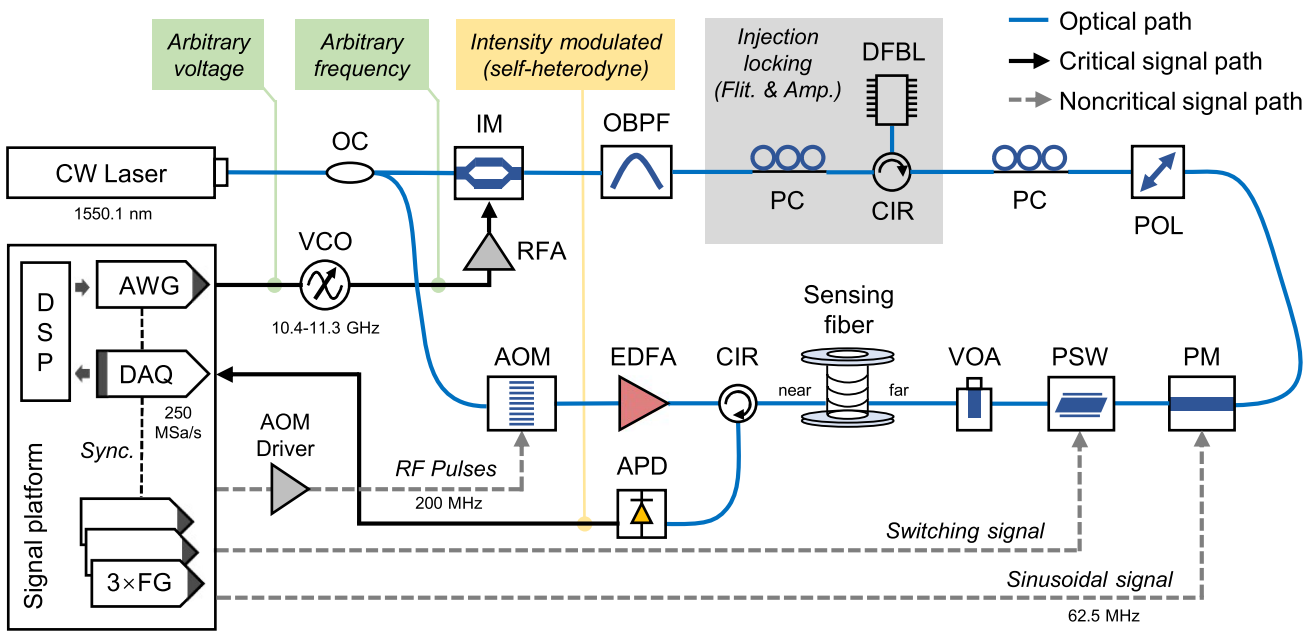

Fig. 4. The experimental setup of the DPDH BOTDA system. OC: optical coupler; IM: intensity modulator; OBPF: optical band-pass filter; PC: polarization controller; DFBL: distributed feedback laser; POL: optical polarizer; AOM: acousto-optic modulator; EDFA: erbium-doped fiber amplifier; CIR: optical circulator; VOA: variable optical attenuator; PSW: polarization switch; PM: phase modulator; APD: avalanche photodetector; DSP: digital signal processor; AWG: arbitrary waveform generator; DAQ: data acquisition card; FG: function generator; VCO: voltage-controlled oscillator; RFA: radio-frequency amplifier.

III. The Experimental System

The experimental system of DPDH BOTDA is given in Fig. 4. A continuous-wave (CW) single frequency laser (NKT, Adjustik E-15) operating at 1550.1 nm is the light source, whose output is divided into two branches by an optical coupler (OC). In the lower branch, an acousto-optic modulator (AOM) is used to generate the pulsed pump light, followed by an erbium-doped fiber amplifier (EDFA) boosting the pulses. The AOM also shifts the frequency of pump light upward by $f_{\mathrm{aom}} = 200$ MHz. The pump pulses enter the sensing fiber from the near end through an optical circulator (CIR), then stimulates the backward Brillouin backscattering with the CW probe light (will be described later) from the far end of the sensing fiber. The response light after SBS is guided by the CIR again into an avalanche photodetector (APD, Thorlabs APD430 C).

The upper branch in the system, which constructs the CW probe light, consists of an intensity modulator (IM, Fujitsu F74 M) for ~11 GHz sideband generation, an optical filter subsystem, a phase modulator (PM, iXblue MPZ-LN-10) for the PDH error, a polarization switch (PSW, iXblue PS-LN-0.1) handling polarization fading effect, and a variable optical attenuator (VOA). The IM is driven by a voltage-controlled oscillator (VCO, ADI HMC1164) whose output frequency range is 10.4 GHz to 11.3 GHz together with an RF amplifier (RFA). The optical filter subsystem picks out the negative first-order sideband as the CW probe (Stokes) light. It contains an optical band-pass filter (OBPF, Santec OTF-970) as first stage filter, an isolator-removed distributed feedback laser (DFBL) combined with CIR and polarization controller (PC) performing injection locking as second stage filter and amplifier [31], and another PC and an optical polarizer (POL) to recover a slow-axis only output. The PM applies $f_{\mathrm{M}} = 62.5$ MHz sinusoidal phase modulation on the probe light, with modulation depth $\beta = 1.1$ ($P_{\mathrm{isb}}/P_{\mathrm{ica}} = 0.4$) to achieve a maximum frequency discriminant [2]. The PSW switches the probe light alternatively into two orthogonal polarization states between adjacent interrogations to compensate the polarization fading effect. The usage of PSW has a benefit that the compensation can be finished in two interrogations [32], rather than tens to hundreds of times at least by a random polarization scrambler to achieve acceptable mitigation, which could potentially restricts the measurement speed. The VOA adjusts the power of probe light which enters the sensing fiber from the far end.

The system control is realized by a signal platform performing signal generation, acquisition, and processing. An arbitrary waveform generator (AWG, NI PXIe-5451) module generates the voltage signal tuning the VCO. A data acquisition card (DAQ, NI PXIe-5171) acquires the output signal of the APD, with sampling rate of 250 MSa/s which is exactly 4 times of the modulation frequency of the PM. The AWG and the DAQ cooperate with a computer (NI PXIe-8135) via high-speed data link for data streaming, and the computer acts as a digital signal processing (DSP) unit in the system. In addition, some function generator (FG, NI PXIe-5422) modules generate signals driving the AOM (pulsed signal), the PSW (switching signal), and the PM (sinusoidal signal). The AWG, DAQ and FGs should be properly synchronized in clocking and triggering for valid cooperation.

The DPDH BOTDA scheme is implemented based on the experimental system above. Since the distributed form feedback control is an iterative operation, we call the process that

the system finishing one BFS distribution measurement and adjusting the arbitrarily accessing frequency $\nu'(z)$ once, i.e., the granularity of distributed form feedback control, an *iteration*. And in order to achieve an adequate SNR, one iteration can contain a certain number of *interrogations* for averaging. We denote the arbitrary distance-varying frequency difference between pump and probe to be used in the k-th iteration as $\nu'_{\{k\}}(z)$ with $k = 0, 1, \cdots$ and $z \in [0, L]$. The time $t = 0$ is defined as the time that a pump pulse is entering the near end of sensing fiber in each interrogation.

To accomplish the interrogations during the k-th iteration, the output voltage signal of the AWG which controls the VCO to compose the probe light should be

$$V_{\text{awg}}^{\{k\}}(t) = \mathcal{V}\left\{\nu'_{\{k\}}\left[\frac{c_0}{2n_{\text{g}}}\left(t + \frac{n_{\text{g}}L_{\text{f}}}{c_0}\right)\right] - f_{\text{aom}}\right\},$$
$$t \in \left[-\frac{n_{\text{g}}L_{\text{f}}}{c_0}, \frac{n_{\text{g}}(2L - L_{\text{f}})}{c_0}\right] \tag{10}$$

where $\mathcal{V}(\cdot)$ represents the voltage vs. frequency curve of the VCO, usually a 2 to 3 order polynomial function, L_{f} is the actual length of sensing fiber, and the time-of-flight principle $z = c_0 t/2n_{\text{g}}$ is employed. Duration of the signal is identical to the round-trip time of light in fiber, while the term $(t + n_{\text{g}}L_{\text{f}}/c_0)$ in (10) indicates that the signal should be one-way flight time ahead of $t = 0$ because the probe light enters the fiber from its far end, which has been well illustrated and explained in [12], [27]. It is worth mentioning that such a time leading should be calculated by the actual fiber length L_{f} rather than the nominal maximum distance of BOTDA L. The frequency shift introduced by the AOM f_{aom} in the pump branch should also be taken into account.

The averaged received signal in the k-th iteration can be expressed as

$$\overline{V}_{\text{pd}}^{\{k\}}(t) = \frac{1}{N}\sum_{i,\xi} V_{\text{pd}}^{\{k,i,\xi\}}(t),$$
$$t \in \left[0, \frac{2n_{\text{g}}L}{c_0}\right], \quad i = 0, 1, \ldots, N-1, \quad \xi = \perp, \| \tag{11}$$

where $V_{\text{pd}}^{\{k,i,\xi\}}(t)$ represents the output signal of APD in each repeated interrogation (denoted by i) under two orthogonal polarization states (denoted by ξ), and the total interrogation count is $2N$. The denominator N (rather than $2N$) in (11) is because the sum (rather than the average) of two polarization states corresponds to the fading free situation in (4) [32]. Then, the PDH error can be calculated by

$$\varepsilon^{\{k\}}(z) = C^{-1} \cdot \text{LPF}[\overline{V}_{\text{pd}}^{\{k\}}(t) \cdot 2\cos(2\pi f_{\text{m}} t + \phi)],$$
$$C = \frac{G_{\text{t}} R_{\text{p}} A_{\text{eff}} n \varepsilon_0 c_0}{2}, \quad z = c_0 t/2n_{\text{g}} \in [0, L] \tag{12}$$

where LPF$(\cdot)$ represents digital low pass filtering, G_{t} and R_{p} are trans-impedance gain and responsivity of the PD respectively, n is the refractive index of fiber, and ε_0 is the dielectric constant. The cut-off frequency of the LPF must be equal to or higher than the reciprocal of pump pulse duration t_{p}^{-1} to guarantee a desired spatial resolution. The rescaling by C^{-1} made in (12) is to keep an equivalence in amplitude between the acquired PDH error and the theoretical frequency discriminant in (9) where the conceptual relative power $P = |E|^2$ is taken.

With the error signal $\varepsilon^{\{k\}}(z)$, the arbitrarily accessing frequency adjustment after the k-th iteration can be performed as

$$\nu'_{\{k+1\}}(z) = \nu'_{\{k\}}(z) - D^{-1}(z)\varepsilon^{\{k\}}(z) \tag{13}$$

which accomplishes the distributed form feedback control, and the resulting $\nu'_{\{k+1\}}(z)$ is to be used in the next iteration immediately. The initial state $\nu'_{\{0\}}(z)$ can be obtained by performing a coarse frequency scanning before the major operation. The iterative process runs continuously, sequentially updating the $\nu'_{\{k\}}(z)$, which is the measurement result of DPDH BOTDA, i.e., the BFS distribution along the sensing fiber over time. Some supplementary details about the suppression of residual amplitude modulation (RAM) effect of PM will be given in later section.

IV. Results and Discussion

A. Performance of the DPDH BOTDA System

In the demonstration experiment, we used G.652.D single mode fiber as sensing fiber. The fiber attenuation is $\alpha = 0.18$ dB/km, and the actual fiber length was $L_{\text{f}} = 24.89$ km. Nominal fiber length L was set to 25.5 km corresponding to a round-trip time of 0.250 ms. The repetition rate of the pump pulses was set to 4.0 kHz. Pulse duration of pump light was $t_{\text{p}} = 50$ ns, corresponding to $l = 5$ m which is the nominal spatial resolution of the system. Power of the pump and probe light entering the sensing fiber were 16 dBm (in peak value) and -9 dBm, respectively. In each iteration, $2N = 400$ times of interrogations were performed. The LPF in (8) is an 8-tap finite-impulse-response (FIR) digital filter, and the adjustable phase shift ϕ is set to 85°. Fig. 5(a) illustrates the PDH error surface of the sensing fiber by performing a full frequency scanning from 10.6 GHz to 10.9 GHz, corresponding to the theoretical result in (8). Note that such a frequency scanning process consuming tens of seconds is only for an overall view of the sensing fiber as a preparation stage.

The DPDH BOTDA system executed the iteration process described in (10) to (13) cyclically, accomplishing the fast BFS distribution measurement. The time cost per iteration was 0.24 s. The initial state $\nu'_{\{0\}}(z)$ can be determined by a zero-crossing searching algorithm from the scanned error curve, which need not be strictly precise since the locking status can be reached after several iteration steps. During the measurement, a positioning stage was used to apply a linearly varying strain from 220 $\mu\varepsilon$ to 2200 $\mu\varepsilon$ on the far end of sensing fiber (24.861 km to 24.879 km). The corresponding measurement result is depicted in Fig. 5(b), where BFS variation from 10.732 GHz to 10.832 GHz within the stretched segment of fiber is clearly observed. Fig. 5(c) shows the BFS curves over distance of the stretching section. The actual spatial resolution of the system is 5.0 m according to the lengths

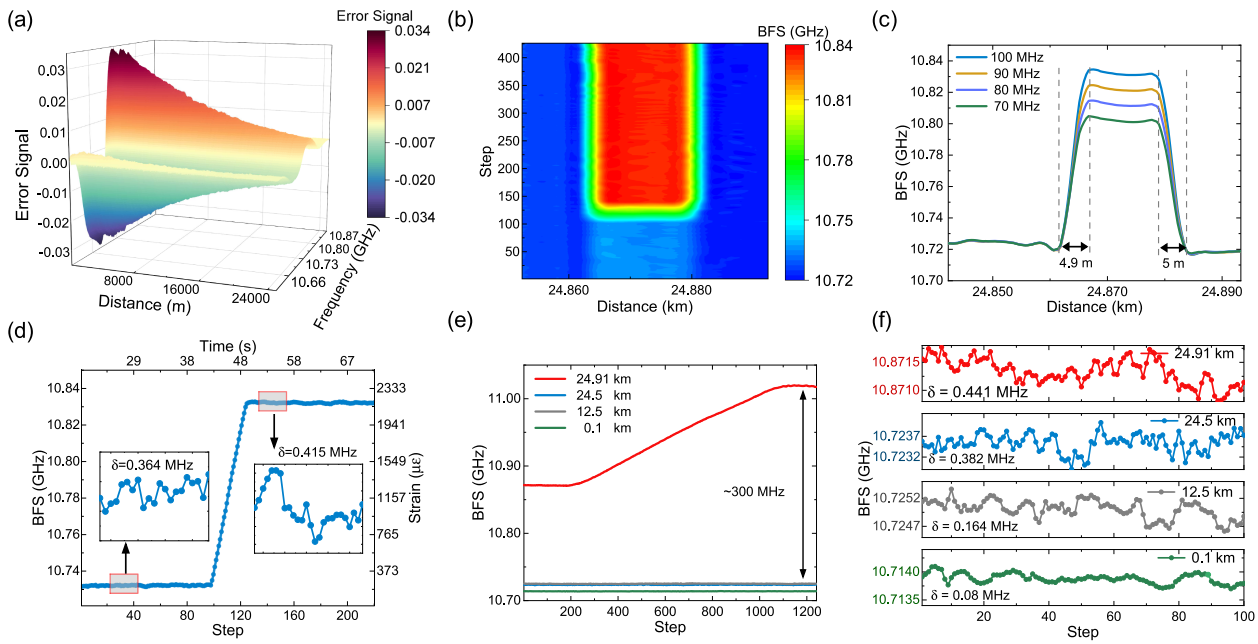

Fig. 5. Results of the DPDH BOTDA experiment. (a) PDH error curve measured by a open loop frequency scan from 10.6 to 10.9 GHz. (b) Measured BFS distribution from 24.855 to 24.890 km in 430 iteration steps. Linearly varying strain was applied between 24.864 and 24.880 km during 100th to 125th iteration. (c) Measured BFS over distance. Data of the four curves are identical to results in sub-figure (b) at the 116th, 119th, 122nd, and 125th iteration, respectively. (d) Measured BFS over iteration step (time). The data is identical to results in sub-figure (b) at $z = 24.875$ km. Standard deviations of the segments before/after the stretch (data in the insets) are 0.364 MHz and 0.415 MHz respectively. (e) BFS measurement results at $z = 0.1$ km (green), 12.5 km (gray), 24.5 km (blue), and 24.91 km (red, the PI coated fiber). The PI coated fiber was heated from 80 °C to 230 °C during the 200th to 1100th step. (f) Insets of the results in sub-figure (e) in the beginning 100 steps. Standard deviations of the curves are 0.080 MHz (at 0.1 km), 0.164 MHz (at 12.5 km), 0.382 MHz (at 24.5 km), and 0.441 MHz (at 25.91 km, the PI coated fiber), respectively.

of rising and falling ranges in the curves, and crosstalk is not observed in adjacent channels. Fig. 5(d) shows the measured BFS over iteration steps (time) at $z = 24.870$ km The rising section in the curve manifests good linearity, and the strain sensitivity is calculated to be 0.0505 MHz/$\mu\varepsilon$. Standard deviations of the sections before and after the stretching are 0.364 MHz and 0.415 MHz respectively, indicating that the sensing resolution at the far end of fiber is 0.4 MHz (corresponding to 8 $\mu\varepsilon$ in strain or 0.4 °C in temperature). In addition, the situation that the sensing resolution would deteriorate when the actual BFS changes over a range of 100 MHz, as occurred in the BGS shape manipulation based measurement range enhancing fast BOTDA schemes [20], [21], [22], does not happen in the DPDH BOTDA system.

To further demonstrate and evaluate the measurement range of the proposed scheme, we appended 40 meters of polyimide (PI) coated fiber at the far end (z from 24.890 km to 24.930 km). The PI coated fiber has higher temperature tolerance, and its intrinsic BFS is ∼100 MHz higher than the G.652.D fiber. This segment of fiber was heated from 80 °C to 230 °C during the measurement, and the measured BFS over iteration step at $z =$ 0.1 km, 12.5 km, 24.5 km and 24.91 km (the PI coated fiber) are shown in Fig. 5(e) and (f). In Fig. 5(e), BFS readouts covering a range of 300 MHz indicates that the sensing system supports a large measurement range which is irrelevant to the width of BGS. Sensing resolutions (standard deviation of 100 samples) at 0.1 km (start), 12.5 km (middle), and 24.9 km (end) are calculated to be 0.08 MHz, 0.16 MHz, and 0.4 MHz respectively, as given in Fig. 5(f).

B. Tolerance to Non-Ideal Effects and the Robustness

The non-local effect [29] is one of the major detrimental effects in BOTDA systems. It refers to the potentially inconsistent power depletion of the pump wave during propagation in fiber, due to the probe wave with varying frequency and the arbitrary BFS distribution. Such effect can distort the desired SBS response in intensity, especially at the far end of long sensing fiber, hence leads to systematic errors [29]. For BOTDA schemes susceptible to the non-local effect, such as frequency-scanning BOTDAs, the solution is to conservatively select a low probe power, but the system SNR would decrease to some degree. While, the DPDH BOTDA scheme is tolerant to the non-local effect, because the fluctuation of pump power never breaks the relationship that a zero PDH error always corresponds to the center of BGS, which is an advantage of BOTDA schemes employing phase-modulated probe light and self-heterodyne detection [16].

To verify this, a comparative experiment of the DPDH BOTDA and frequency-scanning BOTDA was conducted, based on the same experiment system but different control programs. The last tens of meters of sensing fiber ($z \geq 24.82$ km) was heated to about 10 °C higher than the ambient temperature (to introduce ∼ 10 MHz BFS change), which is the configuration

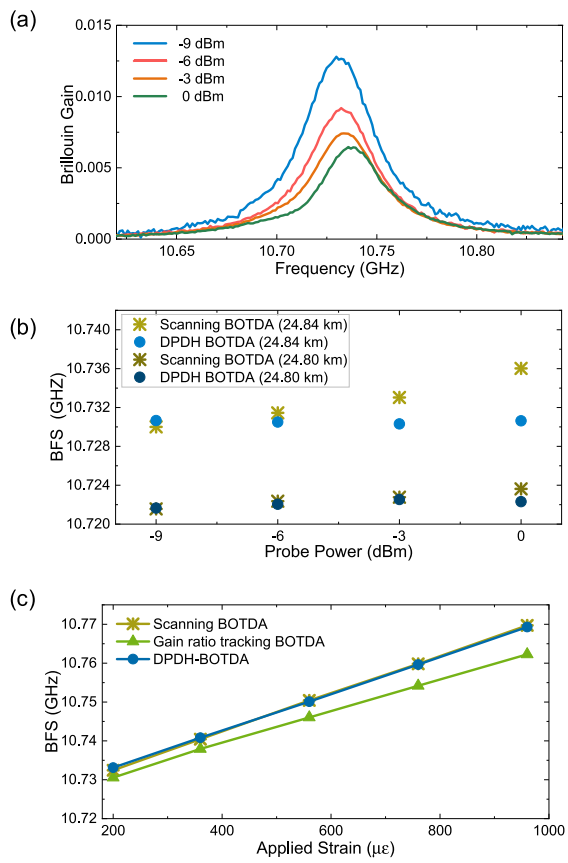

Fig. 6. (a) BGS curves measured by frequency-scanning BOTDA to demonstrate the non-local effect, with probe power of -9 dBm, -6 dBm, -3 dBm, and 0 dBm, respectively. (b) Measured BFS results at $z = 24.84$ km (with light colors) and $z = 24.80$ km (with dark colors) by DPDH BOTDA (in circles) and scanning BOTDA (in stars). BFS values of $z = 24.84$ km, scanning BOTDA are central frequencies of spectra in sub-figure (a). (c) Measured BFS when 200 $\mu\varepsilon$, 380 $\mu\varepsilon$, 580 $\mu\varepsilon$, 780 $\mu\varepsilon$, or 980 $\mu\varepsilon$ strain was applied, by scanning BOTDA (in stars), DPDH BOTDA (in circles) and gain ratio BOTDA (in triangles) for comparison. Here the scanning BOTDA is *not* affected by the non-local effect and gives the true results.

demonstrating a significant non-local effect according to [29]. BGS curves measured by the frequency-scanning BOTDA in the heated section ($z = 24.84$ km) with probe power of -9 dBm, -6 dBm, -3 dBm and 0 dBm are shown in Fig. 6(a). It can be observed that as the probe power increases and exceeds a threshold value of -6 dBm, the central peak of the curve apparently moves right, which is the undesirable distortion caused by non-local effect. By contrast, as shown in Fig. 6(b), the BFS results measured by the DPDH BOTDA at $z = 24.84$ km under different probe power are consistent, indicating that the DPDH BOTDA has an ability to avoid the impact of the non-local effect.

Besides of the principal non-local effect, the DPDH BOTDA is also tolerant to engineerical laser power fluctuations. It has been mentioned that the SA BOTDA [12] and CC BOTDA [27] are sensitive to power fluctuation of pump and probe light, and the closed-loop gain-ratio tracking BOTDA was proposed subsequently to cancel the impact of power fluctuations [28]. However, when we compared the performance of the DPDH BOTDA and the gain-ratio tracking BOTDA, it was found that the former has a more complete tolerance to the power fluctuations. Fig. 6(c) gives the BFS results when strain from 200 $\mu\varepsilon$ to 1000 $\mu\varepsilon$ was applied, measured by frequency-scanning BOTDA, DPDH BOTDA or gain-ratio tracking BOTDA separately. These experiments for comparison were also based on the system in Fig. 4, except that the part of injection locking was removed to match the hardware configuration in [28]. Results of the DPDH BOTDA are consistent well with the scanning BOTDA (the real situation), but deviations about 3 to 10 MHz can be observed in the results of gain-ratio tracking BOTDA. The deviation is found to be caused by the uncertainty of IM insertion loss over modulation frequency (and it also slowly drifts with ambient temperature change), which leads to unbalanced probe powers in the dual frequency interrogations, and hence breaks the uniformity in converting the absolute PD received optical intensities to the relative Brillouin gains. For the gain-ratio tracking BOTDA where a pair of probes with arbitrarily varying frequency is used, it is complicated to conduct a fine, in-situ compensation. While, such engineerical frequency domain power fluctuation is not a concern of the DPDH BOTDA because the PDH error is obtained from *single* self-heterodyne (self-referenced) probe light, even if with arbitrary frequency.

Phase modulation is the key to obtain the elegant PDH error. Therefore, the undesired residual amplitude modulation (RAM) effect of phase modulators [33], is one of the major concerns in PDH systems [34] and has been well studied. Owing to the existence of parasitic reflected waves and other non-idealities in the crystal, the output light of PM carries unwanted amplitude fluctuation with the same frequency of the applied voltage. Such RAM effect breaks the conjugation relationship between the positive and negative first order sidebands, and hence the PDH error curve distorts, leading to a systematic error (bias) in frequency discrimination. In our experiment, we used the tailing segment of PD output signal where z exceeds the actual length of sensing fiber (as illustrated in Fig. 7(a)) to get the RAM status and compensate it. The amplitude fluctuation with frequency of f_{m} in the segment is completely caused by the RAM effect because the SBS no longer exists. This segment of signal participates the calculation in (12), resulting a parasitic PDH error $\varepsilon_{\mathrm{ram}}$ which quantitatively characterizes the RAM. By subtracting the $\varepsilon_{\mathrm{ram}}$ from $\varepsilon(z)$ in each iteration, the impact of RAM effect can be compensated. In order to explain the compensation method more clearly, we performed a frequency scan and show the corresponding results in Fig. 7(b). The curve in light green is the original PDH error which is affected by the RAM effect and hence manifests a negative bias. The brown curve is the calculated parasitic PDH error mentioned above. Their difference, i.e., the curve in blue, is the compensated PDH error in which the bias no longer appears, indicating that the impact of RAM effect is compensated. Since it is a conventional configuration in distributed optical fiber sensors that the nominal maximum distance is slightly larger than the actual fiber length, the method above does not require any additional hardware. Besides of the method above, approaches to suppress the RAM effect in PDH systems by passive isolation [35] and active feedback [34], [36] are also available.

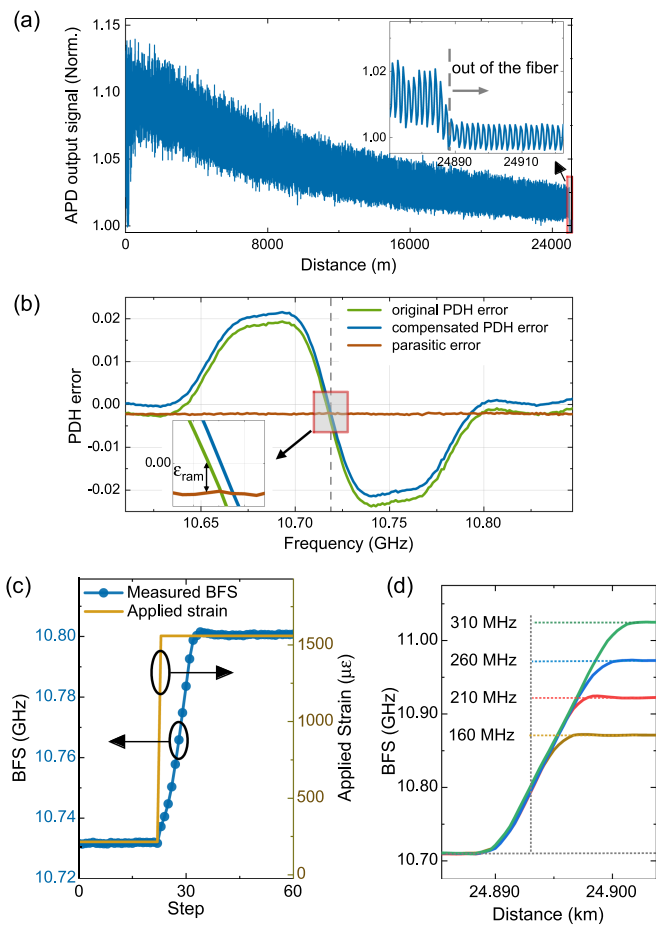

Fig. 7. (a) Acquired APD output signal during one interrogation. Horizontal axis is converted into distance by the time-of-flight principle. Signal after 24.88 km (in the inset) manifests an intensity modulation merely due to the RAM effect. (b) Original PDH error (green), parasitic error due to RAM (brown), and error after compensation (blue) in the same frequency scan. (c) BFS measurement result over iteration step (time) when a sudden, large strain change of 1.4 mε occurs. (d) Measured BFS over distance in the region where a large BFS slide of 160 MHz, 210 MHz, 260 MHz, or 310 MHz actually exists.

The risk of locking failure is a common issue for closed loop systems. For the DPDH BOTDA system, this issue is worthy to be investigated, considering that the iteration rate is not very fast and restricted by the product of average times and round trip time of light in long sensing fiber. In the experiment, we set the interrogation times per iteration to 400 to achieve a relatively high SNR for the 25 km fiber, and the iteration speed is around 0.24 s per step in actual. From Fig. 7(b), it can be seen that the linear section centered at BFS point in the error curve covers about ± 12 MHz range (roughly equals to $\pm\Delta_{\mathrm{B}}/2$). Within the linear section, frequency discrimination by the proportionality coefficient $D(z)$ is correct, which ensures the iteration result in (13) to track the real BFS. So, the maximum tracking rate in our experiment is 50 MHz/s, corresponding to 50 °C/s in temperature or 1000 $\mu\varepsilon$/s in strain, which is sufficient to support general situations in application. The effective range of the error signal is about ± 70 MHz (approximately the same to f_{m}), which is the maximum frequency mismatch tolerance of the system. Fig. 7(c) shows the measurement result of an immediate 1.4 mε strain signal. Although the system can not track such a strong signal during the stretching, it still returns to a locked state after several iterations.

As a *distributed* fiber optic sensor, the system's tracking ability along distance z is also a concern. The maximum tuning speed of VCO is the major restriction, which in our experiment is 3.2 GHz/μs, corresponding to 32 MHz/m by the time-of-flight principle. Therefore, the system can only track BFS changes up to 160 MHz across a distance interval of 5 m (a spatial resolution). If BFS step along the fiber exceeding this value occurs, there will be a degradation in spatial resolution, as illustrated in Fig. 7(d). Such restriction can be relaxed by further increasing the frequency slew rate, e.g., optimizing the impedance of AWG-VCO link, or using VCO with higher frequency tuning sensitivity.

C. Discussion

Works in [23], [24], [25], [26] contributed scanning-free fast BOTDA schemes based on rapidly-repeated chirping or frequency comb probe light achieving excellent performance. From a generalized perspective, there is a technical similarity between these schemes and the DPDH technique in this work, which is the usage of probe light under fast, periodically modulation, bringing a monodramic (i.e., single-valued or well-posed) BFS discrimination, from each segment of the PD received signal within a spatial resolution. One advantage of the PDH technique employing sinusoidal phase modulated probe light lies in the convenience of frequency discrimination, by simply multiplying (frequency mixing) and then filtering (low pass) the received signal. By contrast, the BFS obtainment with chirping or frequency comb probe light takes considerable computation time, which involves massive nonlinear fitting or peak searching tasks, and hence might affect the measurement speed in some applications requiring instant readout in long-term continuous observation.

Besides, in the authors opinion, there would be a trade off between the measurement range and the SNR when using chirping or frequency comb probe light covering a large band (such as several hundreds of megahertz) to support an adequate measurement range. The reason is that the shot noise from PD is persistent and determined by the total power of the probe light, while only a part of probe light within the BGS bandwidth (30 MHz for communication fiber) simulates the Brillouin scattering light which carries valid information. So, the concept in the DPDH BOTDA scheme that using probe light with arbitrary self-adapted *center* frequency, instead of superfluous fully-covered probe, could be also helpful to relax the trade off mentioned above.

In the demonstration experiment, we chose the widely studied gain-type configuration in which energy transfers from the pump pulse to the Stokes probe light. Since results in Fig. 6 manifested a high tolerance to the pump power fluctuation, it is reasonable to infer that the proposed method could be effective when combined with the loss-type (energy transfers from the anti-Stokes probe light to the pump pulse) [37] or dual-sideband probe BOTDA schemes [38] where pump power depletion in

long fiber can be mitigated but the performance is still affected by BGS shape distortion [39], thereby further improving the maximum measurement distance of the DPDH BOTDA scheme. Besides, during the experiment we found that polarization fading was not quite perfectly removed, leading to random intensity drift of the PD output signal. Such random drift has a vague cut-off frequency of 20 MHz (equaling to the reciprocal of the pump pulse duration 50 ns), and it can be partially suppressed by the single-frequency synchronous demodulation (62.5 MHz in our experiment), but still affects the BFS resolution to some degree. Therefore, optimizations on handling the polarization fading effect could be meaningful to further improve the sensing resolution. Another approach is to use higher phase modulation frequency, thus up-shifting the demodulation bandwidth apart from the polarization fading cut-off frequency. And for high-bandwidth demodulation, introducing a dedicated circuit (more discussion will be given later) could be convenient.

In the aspect of the engineering of the DPDH BOTDA system, a primary consideration is the optimization of iteration rate, which determines not only the system's measurement speed, but also the BFS tracking ability. In our experiment, the theoretical lower bound of time cost per iteration is 0.10 s (400 $\times$ 0.25 ms), but it was 0.24 s in actual which is mainly due to the limitation of data transfer speed from DAQ to computer and can be optimized in future. While, the computation time cost is not a major bottle neck. In our system, the interrogation, data transfer, and calculation were organized in a pipelined pattern. So, the dense averaging calculation in (11) shares the time of light round tripping in fiber during the repeated interrogation. Besides, calculations in (10), (12) and (13) are convenient $O(n)$ algorithms and performed only once per iteration.In sum, proper optimizations on the realization of the controller, in order to approach the theoretical iteration rate, are beneficial to the sensing system.

The synchronous demodulation in (12) can also be realized by using hardware frequency mixer and LPF circuit or locked-in amplifiers for sure, as did in most PDH loops and in [15], [16]. Here, it should be noted that the time constant of the hardware filter must not be greater than the pump pulse duration to guarantee the desired spatial resolution. Fast BOTDA schemes in [15], [16] employing phase-modulated probe light used the +1st order sideband to interrogate the BGS, and good performance has been achieved. And in the authors' opinion, using the carrier to align with the BGS (as in the PDH technique) is also available, and has a more strong self-heterodyne which could further improve the SNR. As for the signal generation part, the combination of AWG (140 MHz, 400 MSa/s in our experement) and VCO relaxes the bandwidth requirement in an AWG only configuration, which requires the AWG to have at least hundreds of megahertz bandwidth and Giga level sampling rate to support a sufficient measurement range. If the potential systematic error due to VCO thermal/pushing drift is concerned, an RF counter in parallel can be introduced to perform in-situ calibration of the VCO.

As pointed out in [27], [28], multiplying a scaling factor from 0 to 1 on the feedback coefficient (The term $D^{-1}(z)$ in (13)) establishes a trade-off between sensing resolution and response bandwidth. The authors would suggest to always use the optimal feedback coefficient to achieve a maximum tracking ability first, then introduce low pass filters in post-stage data analysis to suppress the noise if necessary. Lastly, supplementary functions borrowed from existed PDH engineering skills can be introduced into the system. For example, locking failure at any position can be observed by monitoring the strength of 2nd harmonic component in the PD received signal (similar to (12) except using $2f_{\mathrm{m}}$ instead of f_{m}). If any *local* locking failure occurs, the control unit can override the closed-loop mode within the corresponding segment by an open-loop *local* scan (while other positions are in normal closed-loop mode), taking several iterations to search the real *local* BFS and then recover the locking. Including but not limited to the approaches above, the digital signal platform provides adequate flexibility to further enhance the robustness of the DPDH BOTDA scheme when deployed in applications.

V. Conclusion

In summary, we proposed a fast BOTDA scheme by employing the PDH technique and expanding it into a time-domain distributed pattern, named the DPDH BOTDA. In the demonstrational experiment, BFS measurement resolution of 0.4 MHz and measurement range of 300 MHz were achieved without any frequency scanning process, under 25 km communication fiber, 5 m spatial resolution and 400 times of averaging. The scheme also manifests good response linearity, robustness against the non-local effect and the instability of optical power, and the system hardware and algorithm are convenient for instrumentation. The scheme originates from the well known PDH technique, while it satisfies and covers the thoughts contributed by a variety of works on fast BOTDA. In addition, the concept of the DPDH technique could also be helpful to other optical fiber sensing systems with technical similarities such as Brillouin optical correlation domain analysis (BOCDA) [40], distributed forward Brillouin sensors [41] and distributed Brillouin laser sensors [42].

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